

TUPELO, MS, USA

WMO: 723320

Lat: 34.262N

Lon: 88.771W

Elev: 110

StdP: 100.01

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93862

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-4.8	-14.1	1.1	-3.6	-11.8	1.4	-2.1	9.7	10.4	8.8	8.6	3.0	10	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	35.5	24.3	34.3	24.3	33.2	24.1	26.3	32.2	25.7	31.5	25.3	30.9	2.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.0	29.0	24.1	19.3	28.5	23.6	18.7	28.0	82.0	32.1	79.7	31.8	77.6	31.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.5	6.7	-10.8	37.1	2.5	2.1	-12.6	38.7	-14.0	39.9	-15.4	41.1	-17.3	42.6		
			-11.6	27.4	2.5	0.6	-13.4	27.8	-14.8	28.2	-16.2	28.6	-18.0	29.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	5.7	8.1	12.3	17.0	21.6	25.7	27.3	27.1	23.7	17.6	11.3	7.4
	DBStd	8.81	6.11	5.61	5.29	4.29	3.57	2.46	2.03	2.31	3.43	4.59	4.93	5.41
	HDD10.0	459	162	94	37	3	0	0	0	0	0	2	42	118
	HDD18.3	1592	391	289	195	73	14	0	0	4	71	214	341	
	CDD10.0	3060	29	40	109	215	361	471	536	529	412	237	83	37
	CDD18.3	1151	1	2	9	34	116	221	277	271	166	48	5	1
	CDH23.3	11346	1	7	74	295	1026	2204	2905	2799	1554	444	33	2
CDH26.7	4884	0	1	8	49	348	989	1357	1344	654	129	4	0	
Wind	WSAvg	2.8	3.3	3.5	3.5	3.3	2.7	2.3	2.0	2.1	2.4	2.5	2.8	3.1
Precipitation	PrecAvg	1454	126	126	139	139	133	121	106	99	95	95	112	163
	PrecMax	2194	271	285	238	368	434	282	247	209	273	252	247	378
	PrecMin	938	45	62	40	53	16	18	31	10	9	0	34	51
	PrecStd	257	58	56	50	68	96	72	54	57	69	67	51	75
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.4	27.9	29.7	33.0	35.8	36.7	38.1	35.4	31.9	26.8	22.8
		MCWB	17.0	17.9	18.6	20.2	22.7	23.7	24.5	24.2	22.7	21.0	18.7	18.7
	2%	DB	19.8	21.9	25.5	28.2	31.4	34.2	35.0	35.9	33.7	29.8	23.6	20.3
		MCWB	16.1	16.6	17.5	19.4	22.0	23.9	25.0	24.6	22.8	20.9	17.4	17.0
	5%	DB	17.7	19.5	23.3	26.6	30.2	33.0	33.9	34.3	32.3	28.0	21.7	18.4
		MCWB	15.0	15.0	16.1	18.8	21.5	23.7	24.9	24.4	22.7	19.6	16.8	15.7
	10%	DB	15.6	17.4	21.2	25.0	28.8	31.8	32.8	32.8	30.6	25.9	19.8	16.3
		MCWB	12.8	13.2	15.2	17.9	21.1	23.4	24.6	24.2	22.4	18.9	15.4	13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	19.3	20.4	22.5	24.8	26.3	27.0	27.2	25.4	24.0	20.8	19.8
		MCDB	20.5	22.6	24.7	27.4	29.2	32.1	32.9	33.9	31.2	27.7	23.4	22.4
	2%	WB	17.1	17.9	18.8	21.1	23.7	25.6	26.4	26.3	24.6	22.8	19.4	18.0
		MCDB	19.1	20.7	23.6	25.7	28.9	31.5	32.3	32.3	30.1	27.4	22.3	19.8
	5%	WB	15.4	16.1	17.6	20.2	22.9	25.0	25.8	25.6	24.0	21.6	17.9	16.3
		MCDB	17.5	18.8	21.8	24.7	28.1	30.7	31.6	31.5	29.2	25.7	20.7	18.0
	10%	WB	13.3	14.3	16.3	19.1	22.1	24.4	25.4	25.1	23.4	20.5	16.4	14.1
		MCDB	15.1	16.6	20.2	23.4	27.0	29.7	31.0	30.7	28.1	24.1	19.2	16.0
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.9	11.8	12.1	11.3	10.7	10.2	10.7	11.7	12.6	11.8	10.0
		MCDBR	11.5	13.1	13.9	13.1	12.7	12.2	11.5	12.6	13.0	14.4	13.6	11.4
		MCWBR	8.5	8.6	7.6	6.2	5.0	4.1	3.7	3.7	4.3	6.3	8.3	8.4
	5% WB	MCDBR	10.0	11.2	11.7	10.4	10.7	10.4	10.3	10.4	10.6	10.7	10.7	9.7
		MCWBR	8.3	8.4	7.5	5.9	4.9	4.3	3.9	3.8	4.3	6.0	7.8	8.0
Clear-Sky Solar Irradiance	taub	0.316	0.323	0.360	0.393	0.426	0.459	0.503	0.506	0.431	0.368	0.338	0.316	
	taud	2.515	2.491	2.414	2.320	2.271	2.195	2.105	2.088	2.289	2.440	2.467	2.526	
	Ebn at Noon	897	927	914	894	863	832	792	782	830	863	860	874	
	Edh at Noon	83	95	111	128	136	146	159	159	124	99	87	78	
All-Sky Solar Radiation	RadAvg	2.51	3.01	4.17	5.31	5.94	6.39	6.10	5.61	4.92	3.92	2.86	2.15	
	RadStd	0.21	0.40	0.38	0.35	0.37	0.44	0.35	0.35	0.41	0.51	0.28	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+79

Nomenclature: See separate page